oduction RAG degrades quietly — and what to measure instead

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1. Executive summary

Retrieval quality degrades non-linearly as the corpus grows past the point where the embedding model was calibrated, and the failure is quiet: recall stays flat while precision collapses in the tail. Teams that measure only average relevance will not see it. The mitigation is unglamorous. Chunk boundaries must respect document structure rather than token counts, and the reranker has to be trained on the same distribution it will serve. Everything else is tuning.

41%
PRECISION DROP

2.3x
TAIL LATENCY

$0.72
COST PER QUERY

2. Where the failure hides

Retrieval quality degrades non-linearly as the corpus grows past the point where the embedding model was calibrated, and the failure is quiet: recall stays flat while precision collapses in the tail. Teams that measure only average relevance will not see it.

- Tail queries carry most of the business value and least of the traffic.
- Chunk boundaries drawn on token counts sever the exact clauses users ask about.
- Average relevance is a lagging indicator; it moves last.

Key finding. Precision in the bottom decile of query frequency fell 41% while mean nDCG moved less than two points.

3. Measured results

Configuration	nDCG@10	Tail precision	p95 latency
Baseline dense	0.612	0.301	410 ms
+ structural chunking	0.634	0.447	425 ms
+ in-domain reranker	0.671	0.588	690 ms
+ query rewriting	0.688	0.601	940 ms

Caveat. Latency is measured on warm caches. Cold-start numbers are roughly 1.8x higher and are reported in Appendix B.

4. Recommendations

1. Instrument tail precision as a first-class SLO. 2. Re-chunk on document structure, not token windows.

3. Train the reranker on the served distribution. *The systems that failed did not fail loudly. They returned plausible documents that were subtly wrong.* The mitigation is unglamorous. Chunk boundaries must respect document structure rather than token counts, and the reranker has to be trained on the same distribution it will serve. Everything else is tuning.